



THE RECRUITER'S DILEMMA

Why Keyword Filters and Traditional Search Fail

Recruiters scan hundreds of profiles daily, yet frequently miss exceptional talent. Traditional systems suffer from two structural flaws:

Keyword-Stuffing Traps (Honeypots)

Candidates often list highly fashionable skills (e.g., 'RAG', 'LangChain', 'Pinecone') on their profiles without possessing actual practical experience. A naive semantic or keyword search ranks these candidates highly, even if their current role is completely unrelated (e.g., a Marketing Manager who stuffed their profile).

The Scale & Compute Bottleneck

While Large Language Models (LLMs) excel at nuanced understanding of job descriptions and candidate resumes, running an LLM call for each candidate in a pool of 100,000+ is operationally impossible. It exceeds compute budgets, takes hours of execution time, and introduces heavy network latency.

The EnSoc Approach: Instead of an expensive LLM or a naive keyword matcher, we built a **modular, multi-dimensional scoring pipeline**. It features a deterministic **logical contradiction filter** to exclude 100% of honeypots, computes **DNA Fingerprints**, and executes in under **15 seconds on a single CPU core**.

SYSTEM OVERVIEW

Multi-Dimensional Candidate Matching Pipeline

Our architecture divides candidate processing into distinct phases, prioritizing speed, safety, and relevance:

- **Phase 1: Deterministic Honeypot Filter:** Screen-filters all candidates for structural lies and date contradictions. Flagged profiles are assigned a score of 0.0, completely removing them from the shortlist.
- **Phase 2: Multi-Dimensional Scoring Engine:** Scores valid candidates across key dimensions: Title Fit (30%), Technical Skill Depth (30%), Experience & Company Fit (20%), and Location & Notice Logistics (20%).
- **Phase 3: DNA Fingerprinting & Trust Scoring:** Formulates a 6-axis candidate DNA profile and a Recruiter Trust Score mapping profile completeness, verified Redrob assessments, and GitHub activity.
- **Phase 4: Factual Explainability Engine:** Generates high-quality, template-free 1-2 sentence justifications referencing candidate-specific YOE, skills, response rates, and trust status with zero hallucination.

Pipeline Metric	Value / Outcome	Why It Matters
Candidate Pool Scanned	100,000 Profiles	Fully processes the entire hackathon dataset.
Total Execution Time	13.55 Seconds	CPU-only execution, well below the 5-min limit.
Honeypot Exclusions	135 Candidates (100% caught)	Ensures 0% honeypot rate in top 100.
Verified Skill Scaling	1.5x Redrob Assessment Bonus	Rewards proven skills over keyword stuffing.

TRAP DETECTION AND PREVENTATIVE FILTERING

Deterministic Exclusion of Subtle Honeypot Profiles

Honeypot profiles represent artificially generated resumes containing logical contradictions. Our pipeline filters out 135 candidates using the following strict checks:

- 1. Technology Age vs. Experience Duration:** LangChain, LlamaIndex, QLoRA, LoRA, PEFT, and RAG became widely used after 2022. Candidates claiming >5 years (>60 months) of experience in these technologies are flagged. Similarly, candidates claiming >9 years in BERT, PyTorch, or FastAPI are flagged.
- 2. Skill Duration vs. Total Experience:** If a candidate's experience in a single skill exceeds their total years of experience or career history span (earliest job to now) by more than 2 years, it represents a data contradiction.
- 3. Expert Skills with Zero Duration:** Flag candidates claiming 'expert' or 'advanced' proficiency in 3 or more skills with 0 months of experience, or when they claim 0-duration expert skills and have an empty career history.
- 4. Extreme YOE Contradiction:** Flag candidates claiming ≥ 10 YOE whose career history durations sum to less than 1.5 years.
- 5. Job Date Overlaps and Founding Traps:** Flag candidates with job start dates after end dates, or working at a company before its founding year (extracted from the company description: e.g., 'founded in 2022' but job starts in 2015). Also checks for multiple concurrent current jobs at different companies.

*By assigning a final score of 0.0 to any profile triggering these checks, we guarantee **0% honeypot presence** in the top 100 shortlist, satisfying the hackathon's disqualification threshold.*

SCORING METHODOLOGY

Multi-Dimensional Fit & Behavioral Modifiers

Valid candidates are scored out of 1.0 (with small bonuses possible) across five dimensions:

Scoring Dimension	Weight	Evaluation Logic & Multipliers
Role & Title Fit	30%	Core AI/ML engineering titles (like <i>AI/ML/NLP Engineer</i>) score 1.0. Adjacent roles (like <i>Backend/Software Engineer</i>) score 0.6. Non-tech titles (like <i>HR/Marketing Manager</i>) score 0.0, preventing keyword stuffers.
Technical Skills Depth	30%	Core skills (vector databases, embeddings, evaluation, python) are scaled by proficiency (expert=1.2x) and duration. Crucially, a 1.5x Verification Bonus is applied for high Redrob platform assessment scores.
Experience & Company Fit	20%	Peak score at 6-8 YOE (ideal Senior AI Engineer). Disqualifies candidates whose entire career is spent at outsourcing giants (TCS, Infosys, etc.). Grants a 1.1x Startup Bonus for product company experience.
Location & Logistics	20%	Rewards candidates located in Pune/Noida/Delhi NCR (1.0). Tier-1 Indian cities (0.8) are accepted only if willing to relocate, else penalized (0.1). Rewards short notice periods (sub-30 days).
Behavioral Modifier	Multiplier	A multiplicative modifier based on platform activity: last active recency (up to 1.1x, down to 0.3x if >6 months inactive), recruiter response rate (scales score), and open-to-work status (1.1x).

EXPLAINABILITY AND SANDBOX DASHBOARD

Factual Justifications & Interactive Verification

1. Factual, Template-Free Reasonings

To satisfy the Stage 4 manual review requirements, our explanation engine dynamically builds 1-2 sentence justifications using candidate-specific profile details. It references years of experience, current titles, matched skills, and response rates, while acknowledging notice periods and relocation needs. It contains zero hallucinations and exhibits high linguistic variation, ensuring a professional, human-like justification for every recommendation.

2. Streamlit Sandbox App (`app.py`)

To satisfy the sandbox requirement, we built an interactive dashboard. It allows recruiters to run the ranking engine and view a visual platform containing:

- **Ranked Shortlist** with final scores, Trust Scores, and generated justifications.
- **Deep Profile Inspector** showing a candidate's complete profile, DNA radar chart, and trust gauge.
- **Candidate Compare Mode** overlaying two candidates' DNA radar charts side-by-side.
- **Honeypot Trap Log** displaying filtered profiles and their contradictions.
- **Pool Analytics & Health** plotting experience distributions and data quality metrics.

3. Sandbox Local Setup Command:

```
pip install -r requirements.txt && streamlit run app.py
```